ESAT PRACTICE TEST SERIES

MATHS 2 + BIOLOGY

LEVEL 1 · STANDARD

Maths 2 + Biology

Mathematics 1 + Mathematics 2 + Biology

Representative ESAT-style breadth with routine and multi-step applications.

MODULE
Mathematics 1

MODULE
Mathematics 2

MODULE
Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions · three verified module keys

COMBINED ANSWER KEYS

Maths 2 + Biology · Level 1 Standard

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Mathematics 2 + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 3 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 E	Q4 C	Q5 D
Q6 D	Q7 E	Q8 A	Q9 A	Q10 D
Q11 D	Q12 C	Q13 B	Q14 A	Q15 B
Q16 E	Q17 E	Q18 E	Q19 D	Q20 E
Q21 D	Q22 D	Q23 C	Q24 D	Q25 D
Q26 A	Q27 C			

Mathematics 2

Engineering Mock Test 3 · preserved Mathematics 2 module

Q	Answer
Q1	C
Q2	A
Q3	C
Q4	C
Q5	C
Q6	B
Q7	D
Q8	E
Q9	D
Q10	F
Q11	D
Q12	B
Q13	A
Q14	D
Q15	E
Q16	G
Q17	D
Q18	G
Q19	C
Q20	C
Q21	C
Q22	G
Q23	E
Q24	D
Q25	A
Q26	B
Q27	A

Biology

Preserved Biology Standard source set

Q	Answer
Q1	D
Q2	C
Q3	C
Q4	G
Q5	C
Q6	F
Q7	E
Q8	F
Q9	C
Q10	D
Q11	B
Q12	E
Q13	B
Q14	D
Q15	C
Q16	E
Q17	E
Q18	H
Q19	B
Q20	C
Q21	E
Q22	D
Q23	B
Q24	B
Q25	C
Q26	F
Q27	E

LEVEL 1 · STANDARD

Mathematics 1

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 3 - Mathematics 1

Answer C

QUESTION 1

There are 8 consecutive integers. If the sum of the first three is 63, what is the largest one?

- **A** 17
- **B** 25
- **C** 27
- **D** 30
- **E** 31

WORKED SOLUTION

Let the first integer be n. Then $n + (n + 1) + (n + 2) = 63$, so $3n + 3 = 63$, giving $n = 20$. The largest of the eight integers is 27.

M1-02

Mock Test 3 - Mathematics 1 | Answer **B**

QUESTION 2

Joshua is 10 years old and his father is 42 years old. After how many years, his father's age is three times Joshua's age?

A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

After t years, $42 + t = 3(10 + t)$. Hence $42 + t = 30 + 3t$, so $t = 6$.

M1-03

Mock Test 3 - Mathematics 1 | Answer **E**

QUESTION 3

N is a 7-digit number consists of 2 or 3. There are more 2s than 3s in N. If N is divisible by both 3 and 4, what is the remainder of N when it is divided by 7?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Let k be the number of 3s. The digit sum is $14 + k$, so divisibility by 3 gives $k = 1$. Divisibility by 4 forces the last two digits to be 32. Thus $N = 2222232$, and $N \equiv 5 \pmod 7$.

M1-04

Mock Test 3 - Mathematics 1

Answer C

QUESTION 4

The preliminary round of a violin competition is divided into four rounds. According to the rule, the average score of a percipient of these four rounds must be not less than 96 points in order to enter the finals. Shaelyn's scores in the first three rounds are 95, 97, and 94. What is the minimum score that she needs to get on the fourth round in order to enter the finals? (Hope Cup Math Competition, China)

A 96

B 97

C 98

D 99

E 100

WORKED SOLUTION

The total score needed is $4 \times 96 = 384$. The first three scores total $95 + 97 + 94 = 286$, so the fourth score must be at least $384 - 286 = 98$.

M1-05 Mock Test 3 - Mathematics 1 **Answer D**

QUESTION 5

A haunted house has 8 windows. In how many ways can George the Ghost enter the house by one window and leave by a different window?

- **A** 28
- **B** 32
- **C** 40
- **D** 56
- **E** 63

WORKED SOLUTION

There are 8 choices for the entry window and then 7 different choices for the exit window, giving $8 \times 7 = 56$.

M1-06 Mock Test 3 - Mathematics 1 **Answer D**

QUESTION 6

Jackie bought a bottle (30 oz) of 90% alcohol from pharmacy to make homemade disinfectant spray. How many ounces of water should he add in order to dilute it into 75% alcohol? (Hint: The amount of alcohol remains the same.)

A 3

B 4

C 5

D 6

E 7

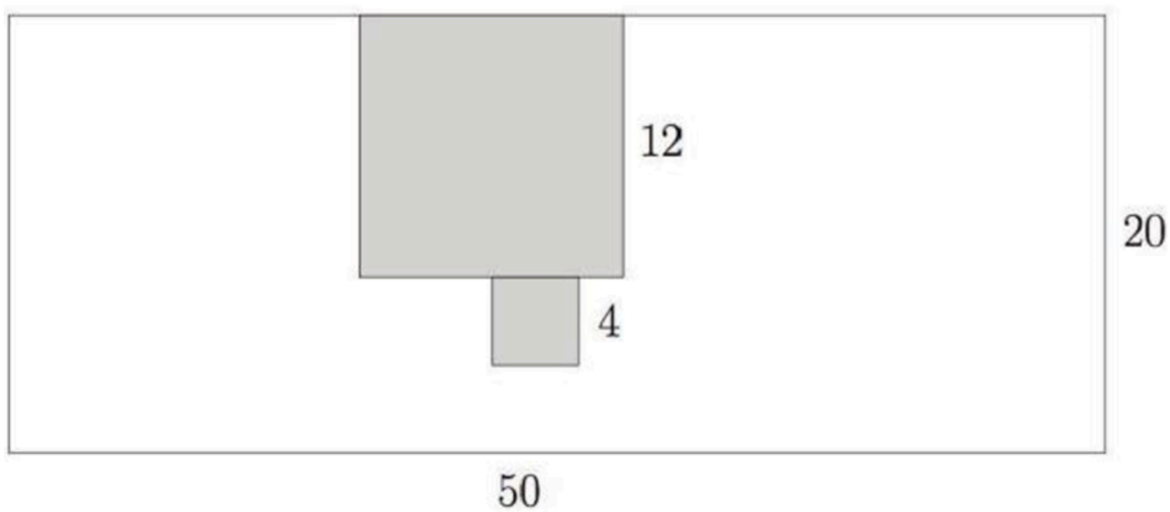
WORKED SOLUTION

The amount of alcohol is $30 \times 0.90 = 27$ oz. If the final concentration is 75%, then $0.75T = 27$, so $T = 36$. Water added: $36 - 30 = 6$ oz.

M1-07 Mock Test 3 - Mathematics 1 Answer E

QUESTION 7

Cut a 12×12 square and a 4×4 square from a 50×20 rectangle paper. What is the perimeter of the remaining part?



A 128

B 140

C 152

D 160

E 172

WORKED SOLUTION

The original rectangle perimeter is $2(50+20)=140$. The 12×12 notch increases the perimeter by 24, and the attached 4×4 cut-out adds another 8. Total perimeter $=140+32=172$.

M1-08 Mock Test 3 - Mathematics 1 Answer A

QUESTION 8

Jasmine keeps adding numbers one by one from the following list

1, 3, 5, 7, ⋯ .

as $1+3=4$, $1+3+5=9$, $1+3+5+7=16$, etc. At one point, she got 379 and realized that one number is missed in the addition. What is the sum of digits of the missing number?

A 3

B 6

C 8

D 9

E 11

WORKED SOLUTION

The sum of the first n odd numbers is n^2. Since 379 is just below $20^2 = 400$, the missing number is $400 - 379 = 21$. The digit sum is $2 + 1 = 3$.

M1-09

Mock Test 3 - Mathematics 1

Answer A

QUESTION 9

$\overline{ab}$ and $\overline{cd}$ are two 2-digit numbers. If

$\overline{ab} - \overline{cd} = 24$

and

$\overline{1ab} - \overline{cd1} = 15,$

what is $a + d$?

A 5

B 8

C 10

D 11

E 12

WORKED SOLUTION

Using place value, $\overline{ab} = 10a + b$ and $\overline{cd} = 10c + d$. Solving the two given equations gives $a + d = 5$.

M1-10

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 10

The speed of a train is 15 meters per second. If it takes the train 10 seconds to pass a lamp post, how long does it take the train to completely pass a 300-meter long tunnel, in seconds? (Hint: first figure out the length of the train based on the information related to the lamppost)

A 10

B 20

C 25

D 30

E 35

WORKED SOLUTION

The train length is $15 \times 10 = 150$ m. To pass the tunnel completely, it travels $300 + 150 = 450$ m. Time $= 450/15 = 30$ s.

M1-11

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 11

How many pairs of positive integers (x, y) satisfy that $\frac{x}{3} = \frac{4}{y}$?

A 3

B 4

C 5

D 6

E 7

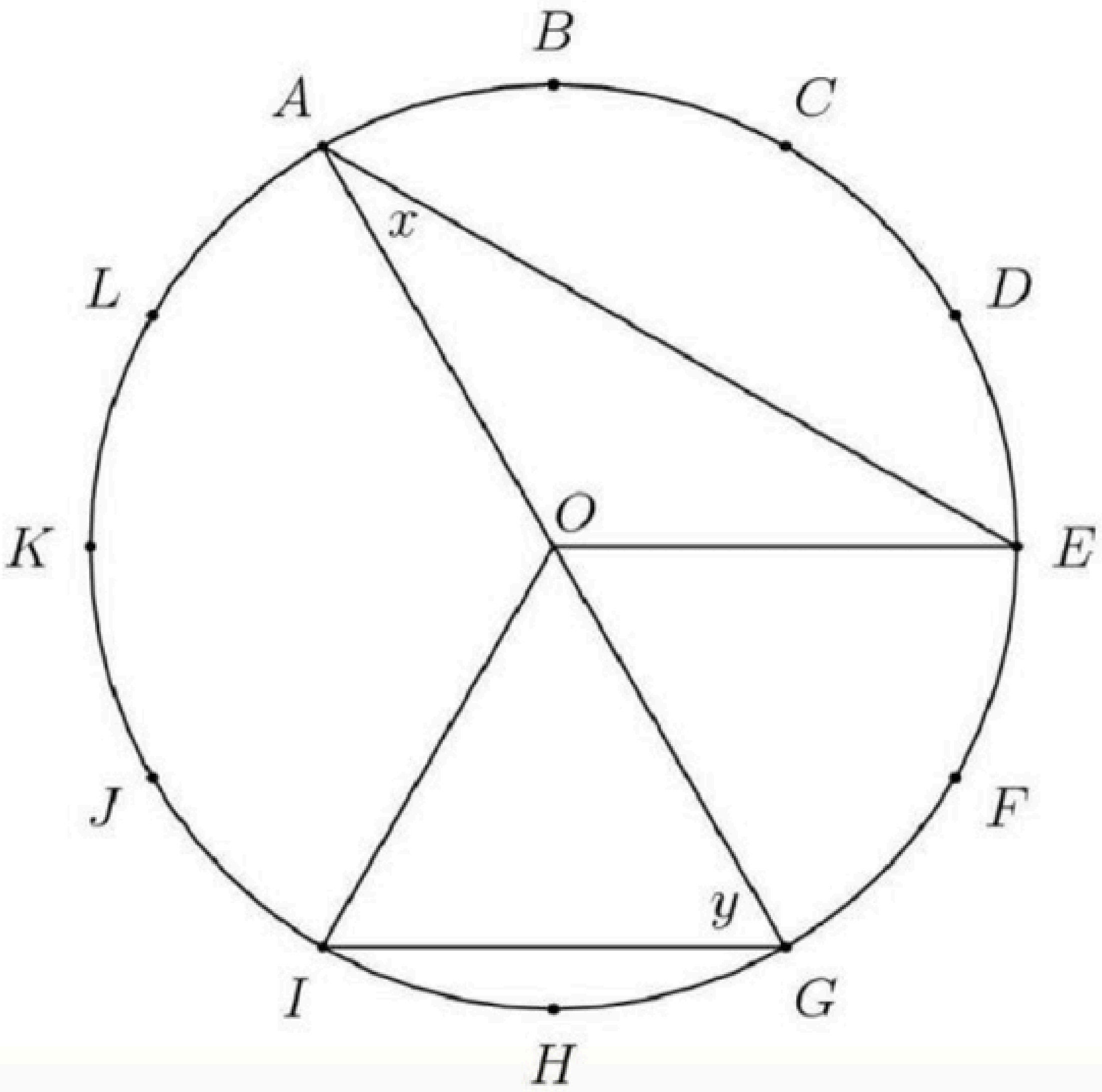
WORKED SOLUTION

From $x/3 = 4/y$, we get $xy = 12$. The positive factor pairs are $(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1)$, so there are 6 pairs.

M1-12 Mock Test 3 - Mathematics 1 Answer **C**

QUESTION 12

The circumference of the circle with center O is divided into 12 equal arcs, marked the letters A through L as seen below. What is the number of degrees in the sum of the angles x and y?



A 75

B 80

C 90

D 120

E 150

WORKED SOLUTION

The circle is divided into 12 equal arcs, so each arc is 30°. Using central and inscribed angle relationships from the diagram gives $x + y = 90^\circ$.

M1-13 Mock Test 3 - Mathematics 1 **Answer B**

QUESTION 13

If for whole numbers x, y and z,

$360 = 2^x \times 3^y \times 5^z$,

what is $x + y + z$? Here $a^n = a \times a \times \cdots \times a$.

For example, $100 = 2 \times 2 \times 5 \times 5 = 2^2 \times 5^2$.

- **A** 5
- **B** 6
- **C** 7
- **D** 8
- **E** 9

WORKED SOLUTION

Prime factorise: $360 = 36 \times 10 = 2^2 \times 3^2 \times 2 \times 5 = 2^3 \times 3^2 \times 5^1$. Hence $x + y + z = 3 + 2 + 1 = 6$.

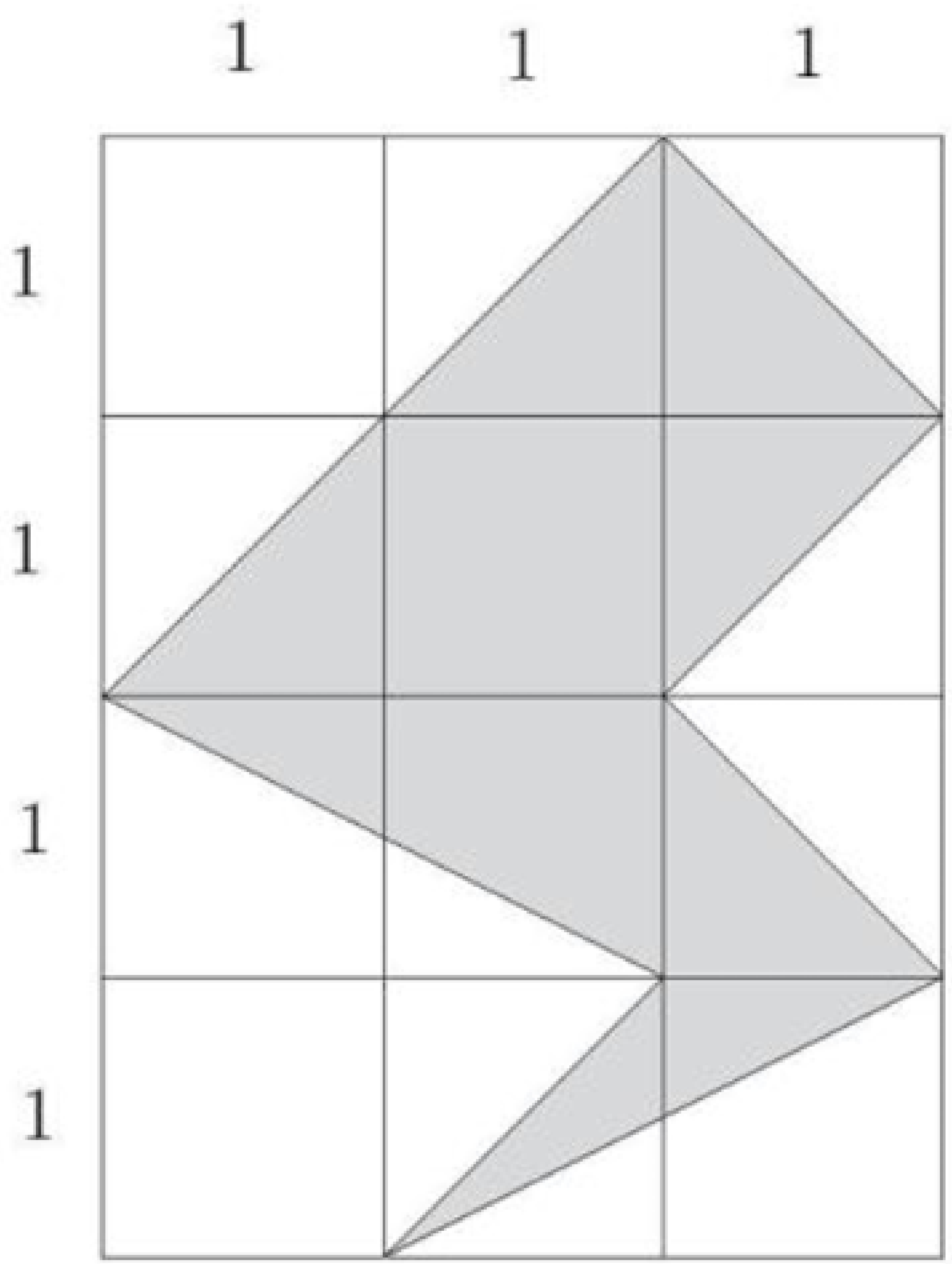
M1-14

Mock Test 3 - Mathematics 1

Answer A

QUESTION 14

What is the area of the shaded region in the following 3×4 grid?



A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

Counting the shaded polygon by subtracting the surrounding right triangles from the 3×4 rectangle gives shaded area 5.

M1-15 Mock Test 3 - Mathematics 1 Answer B

QUESTION 15

Randomly choose 3 different numbers from the set $\{1, 3, 5, 7, 9\}$. What is the probability that the three selected numbers could be side lengths of a triangle?

A $\frac{1}{5}$

B $\frac{3}{10}$

C $\frac{1}{3}$

D $\frac{2}{5}$

E $\frac{1}{2}$

WORKED SOLUTION

There are $\binom{5}{3} = 10$ triples. The valid triangle triples are $(3, 5, 7), (3, 7, 9), (5, 7, 9)$, so the probability is $3/10$.

M1-16

Mock Test 3 - Mathematics 1

Answer E

QUESTION 16

Marley practices exactly one sport each day of the week. She runs three days a week but never on two consecutive days. On Monday she plays basketball and two days later golf. She swims and plays tennis, but she never plays tennis the day after running or swimming. Which day of the week does Marley swim?

- **A** Sunday
- **B** Tuesday
- **C** Thursday
- **D** Friday
- **E** Saturday

WORKED SOLUTION

Monday is basketball and Wednesday is golf. Applying the non-consecutive running rule and the condition that tennis is not immediately after running or swimming leaves Saturday as the swimming day.

M1-17

Mock Test 3 - Mathematics 1

Answer **E**

QUESTION 17

Three toy cars run around a round track. It takes them 8, 10 and 12 seconds to finish one full circle respectively. If three cars start together from the same point and run toward the same direction, after how many seconds will they get back to the starting point together for the first time?

- **A** 20
- **B** 40
- **C** 60
- **D** 80
- **E** 120

WORKED SOLUTION

The cars return together after the least common multiple of $8, 10, 12$, which is 120 seconds.

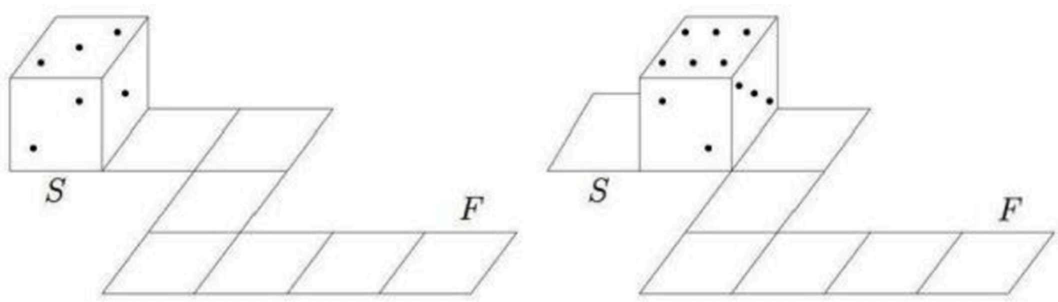
M1-18

Mock Test 3 - Mathematics 1

Answer **E**

QUESTION 18

Opposite faces of a die always add to 7. A die rolls on a circuit as represented below. At the starting point (S) the top face is 3. Which will be the top face at the final point (F)?



A 2

B 3

C 4

D 5

E 6

F 7

G 8

H 9

WORKED SOLUTION

Track the die through the path, using opposite faces summing to 7 after each roll. The final top face is 6.

M1-19

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 19

How many numbers x satisfy that $(x^2 - 2)^2 = 4$?

- **A** 0
- **B** 1
- **C** 2
- **D** 3
- **E** 4
- **F** 5
- **G** 6
- **H** 7

WORKED SOLUTION

$(x^2 - 2)^2 = 4$ gives $x^2 - 2 = \pm 2$. Thus $x^2 = 4$ or $x^2 = 0$, giving $x = -2, 0, 2$. There are 3 real numbers.

M1-20

Mock Test 3 - Mathematics 1

Answer **E**

QUESTION 20

For how many two-digit numbers, if you switch the tens and ones digits (e.g., 18 → 81), the new 2-digit number is 18 more than the original one?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

Let the number be $10a + b$. Reversing digits gives $10b + a$. The condition gives $10b + a = 10a + b + 18$, so $b - a = 2$. There are 7 possible tens digits.

M1-21

Mock Test 3 - Mathematics 1 | Answer D

QUESTION 21

[AMC 8 Problem] Malcolm wants to visit Isabella after school today and knows the street where she lives but doesn't know her house number. She tells him, "My house number has two digits, and exactly three of the following four statements about it are true."

- It is prime.
- It is even.
- It is divisible by 7.
- One of its digits is 9.

This information allows Malcolm to determine Isabella's house number. What is its units digit?

A 4

B 6

C 7

D 8

E 9

WORKED SOLUTION

The number must have exactly three of the four properties. The number 98 is even, divisible by 7, has a digit 9, and is not prime. Its units digit is 8.

M1-22

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 22

In the addition problem shown, m, n, p, and q represent positive digits. When the problem is completed correctly, the value of $m + n + p + q$ is (Canadian Math Competition) t is prime.

t is even.

t is divisible by 7.

)ne of its digits is 9.

information allows N

A 16

B 18

C 22

D 24

E 30

WORKED SOLUTION

From the units column, $3 + 2 + q$ ends in 2, so $q = 7$ with carry 1. From the tens column, $6 + p + 8 + 1$ ends in 4, so $p = 9$. From the hundreds column,

$n+7+5+2$ ends in 0, so $n=6$ with carry $m=2$. Therefore $m+n+p+q=2+6+9+7=24$.

M1-23 Mock Test 3 - Mathematics 1 Answer **C**

QUESTION 23

If two numbers x and y satisfy that $(x+1)^2+(y-5)^2=0$, what is $x+y$?

- **A** −1
- **B** 0
- **C** 4
- **D** 6
- **E** 7

WORKED SOLUTION

A sum of squares is zero only if both squares are zero. Thus $x+1=0$ and $y-5=0$, so $x=-1, y=5$, and $x+y=4$.

M1-24 Mock Test 3 - Mathematics 1 Answer **D**

QUESTION 24

Abel tosses two coins and Ben tosses one coin. What is the probability that they get the equal number of heads?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number all possible cases}}.$$

A $\frac{1}{8}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{3}{8}$

E $\frac{2}{5}$

WORKED SOLUTION

There are 8 equally likely outcomes. Equal heads occurs when Abel gets 0 heads and Ben gets 0 heads, or when Abel gets 1 head and Ben gets 1 head: $1 + 2 = 3$ cases. Probability $= 3/8$.

M1-25 Mock Test 3 - Mathematics 1 Answer **D**

QUESTION 25

x, y, z and w are four numbers. If the average of the following three numbers $x - y,\ y - z,\ z - w$ is 10 and $w = 5$. What is x?

A 15

B 20

C 25

D 35

E there is no enough information

WORKED SOLUTION

The average of $x - y, y - z, z - w$ is 10, so their sum is 30. The sum telescopes: $(x - y) + (y - z) + (z - w) = x - w = 30$. Since $w = 5$, $x = 35$.

M1-26

Mock Test 3 - Mathematics 1

Answer A

QUESTION 26

What is the tens digit of 7^{2020}?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

The last two digits of powers of 7 cycle: $07, 49, 43, 01$. Since 2020 is divisible by 4, 7^{2020} ends in 01, so the tens digit is 0.

M1-27

Mock Test 3 - Mathematics 1

Answer C

QUESTION 27

Two trains 300 miles apart are traveling toward each other along the same track. One train goes 70 miles per hour; the other runs at 80 miles per hour. If both of them leave their stations at 7:00am, when will they meet?

A. 8 : 00 am

B. 8 : 30 am

C. 9 : 00 am

D. 9 : 30 am

E. 10 : 00 am

WORKED SOLUTION

The trains approach each other at $70 + 80 = 150$ miles per hour. Time $= 300/150 = 2$ hours, so they meet at 9:00 am.

LEVEL 1 · STANDARD

Mathematics 2

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Mathematics 2 module. Question wording, option order, answer and solution method are unchanged.

M2-01 Mock Test 3 - Mathematics 2 Answer C

QUESTION 1

Rearrange $y^4 - 4y^3 + 6y^2 - 4y + 2 = x^5 + 7$ to make y the subject, taking the principal fourth-root branch.

A $y = 1 + (x^5 + 7)^{1/4}$

B $y = -1 + (x^5 + 7)^{1/4}$

C $y = 1 + (x^5 + 6)^{1/4}$

D $y = -1 + (x^5 + 6)^{1/4}$

WORKED SOLUTION

Complete the binomial pattern:

$y^4 - 4y^3 + 6y^2 - 4y + 1 = (y - 1)^4$.

So $(y - 1)^4 + 1 = x^5 + 7$, hence $(y - 1)^4 = x^5 + 6$. Taking the principal fourth root gives $y = 1 + (x^5 + 6)^{1/4}$.

M2-02

Mock Test 3 - Mathematics 2

Answer A

QUESTION 2

The aspect ratio of my television screen is 4:3 and the diagonal is 50 inches. What is the area of my television screen?

- **A** 1,200 inches^2
- **B** 1,000 inches^2
- **C** 120 inches^2
- **D** 100 inches^2
- **E** More information needed.

WORKED SOLUTION

Question 2: A

Let the width of the television be $4x$ and the height be $3x$.

By Pythagoras,

$$(4x)^2 + (3x)^2 = 50^2$$

Simplify:

$$25x^2 = 2500 \Rightarrow x = 10$$

Therefore, the screen is 30 inches by 40 inches, so the area is 1200 inches^2.

M2-03

Mock Test 3 - Mathematics 2

Answer C

QUESTION 3

Rearrange the equation $\sqrt{1 + 3x^{-2}} = y^5 + 1$ to make x the subject, taking the positive square-root branch.

A $x = \dfrac{y^{10} + 2y^5}{3}$

B $x = \dfrac{3}{y^{10} + 2y^5}$

C $x = \sqrt{\dfrac{3}{y^{10} + 2y^5}}$

D $x = \sqrt{\dfrac{y^{10} + 2y^5}{3}}$

E $x = \sqrt{\dfrac{y^{10} + 2y^5 + 2}{3}}$

F $x = \sqrt{\dfrac{3}{y^{10} + 2y^5 + 2}}$

WORKED SOLUTION

Square both sides:

$1 + 3x^{-2} = (y^5 + 1)^2 = y^{10} + 2y^5 + 1.$

Thus $3x^{-2} = y^{10} + 2y^5$, so $x^2 = \dfrac{3}{y^{10} + 2y^5}$. Taking the positive branch gives

$x = \sqrt{\dfrac{3}{y^{10} + 2y^5}}$.

M2-04

Mock Test 3 - Mathematics 2

Answer C

QUESTION 4

Solve $3x - 5y = 10$ and $2x + 2y = 13$.

A $(x, y) = \left(\frac{19}{16}, \frac{85}{16}\right)$

B $(x, y) = \left(\frac{85}{16}, -\frac{19}{16}\right)$

C $(x, y) = \left(\frac{85}{16}, \frac{19}{16}\right)$

D $(x, y) = \left(-\frac{85}{16}, -\frac{19}{16}\right)$

E No solutions possible.

WORKED SOLUTION

The easiest way is to double the first equation and triple the second to get:

$6x - 10y = 20$ and $6x + 6y = 39$.

Subtract the first from the second to give: $16y = 19$.

Therefore, $y = \frac{19}{16}$.

Substitute back into the first equation to give $x = \frac{85}{16}$.

M2-05

Mock Test 3 - Mathematics 2

Answer C

QUESTION 5

The two inequalities $x + y \leq 3$ and $x^3 - y^2 < 3$ define a region on a plane. Which of the following points is inside the region?

A (2, 1)

B (2.5, 1)

C (1, 2)

D (3, 5)

E (1, 2.5)

F None of the above.

WORKED SOLUTION

This is fairly straightforward; the first inequality is the easier one to work with: options B, D and E violate it, so we just need to check A and C in the second inequality.

C: $1^3 - 2^2 < 3$, but A: $2^3 - 1^2 > 3$.

M2-06 Mock Test 3 - Mathematics 2 Answer B

QUESTION 6

How many times do $y = x + 4$ and $y = 4x^2 + 5x + 5$ intersect?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Whilst this can be done graphically, it is quicker to do algebraically. Intersections occur where the curves have the same coordinates.

Thus: $x + 4 = 4x^2 + 5x + 5$.

Simplify: $4x^2 + 4x + 1 = 0$.

Factorise: $(2x + 1)(2x + 1) = 0$.

Thus, the two graphs only intersect once at $x = -\frac{1}{2}$.

M2-07

Mock Test 3 - Mathematics 2

Answer D

QUESTION 7

How many times do $y = x^3$ and $y = x$ intersect?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Answer: D

It's better to do this algebraically as the equations are easy to work with and you would need to sketch very accurately to get the answer. Intersections occur where the curves have the same coordinates. Thus: $x^3 = x$.

$$x^3 - x = 0$$

Thus: $x(x^2 - 1) = 0$

Spot the 'difference between two squares': $x(x + 1)(x - 1) = 0$

Thus there are 3 intersections: at $x = 0, 1$ and -1.

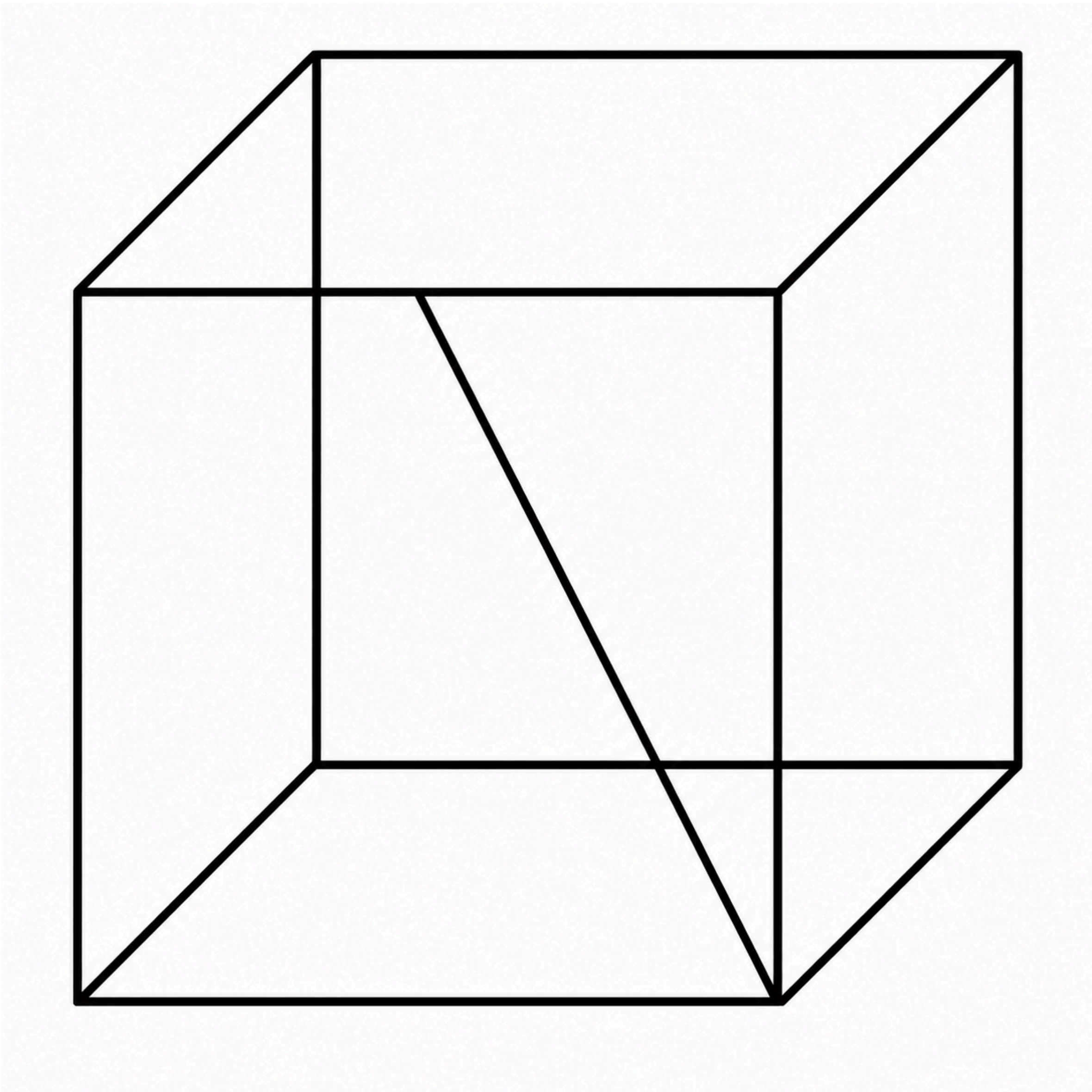
M2-08

Mock Test 3 - Mathematics 2

Answer E

QUESTION 8

A cube has unit length sides. What is the length of a line joining a vertex to the midpoint of the opposite side?



A $\sqrt{2}$

B $\sqrt{\frac{3}{2}}$

C $\sqrt{3}$

D $\sqrt{5}$

E $\frac{\sqrt{5}}{2}$

WORKED SOLUTION

Answer: E

Note that the line is the hypotenuse of a right angled triangle with one side unit length and one side of length $\frac{1}{2}$.

By Pythagoras, $\left(\frac{1}{2}\right)^2 + 1^2 = x^2$.

Thus, $x^2 = \frac{1}{4} + 1 = \frac{5}{4}$.

$x = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$.

M2-09 Mock Test 3 - Mathematics 2 Answer **D**

QUESTION 9

Find the value of

$$\int_1^4 \frac{3 - 2x}{x\sqrt{x}}\, dx$$

A $-\frac{13}{2}$

B $-\frac{85}{16}$

C $-\frac{13}{8}$

D -1

E $-\frac{1}{4}$

F $\frac{7}{4}$

G 7

WORKED SOLUTION

Answer: D

$$\int_1^4 \frac{3-2x}{x\sqrt{x}}\,dx$$

(in terms of powers of x and split up the fraction)

$$= \int_1^4 \left(\frac{3}{x^{3/2}} - \frac{2}{x^{1/2}}\right) dx = \int_1^4 (3x^{-3/2} - 2x^{-1/2})\,dx$$

$$= \left[\frac{3x^{-1/2}}{-1/2} - \frac{2x^{1/2}}{1/2}\right]_1^4$$

$$= [-6x^{-1/2} - 4x^{1/2}]_1^4$$

$$= (-6 \cdot \tfrac{1}{2} - 4 \cdot 2) - (-6 \cdot 1 - 4 \cdot 1)$$

$$= -11 - (-10)$$

$$= -1$$

M2-10

Mock Test 3 - Mathematics 2

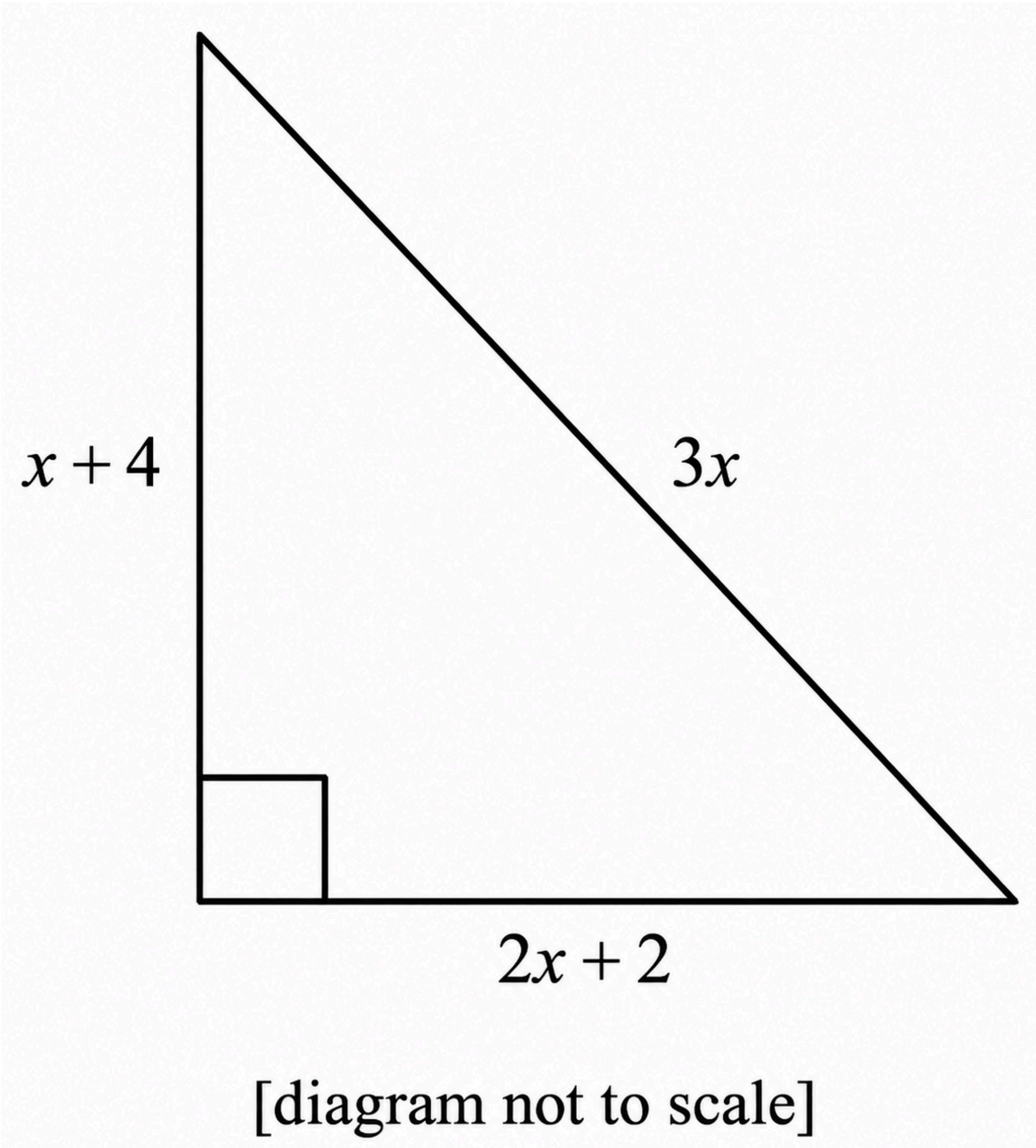
Answer F

QUESTION 10

The diagram shows a right-angled triangle, with sides of length $x + 4$, $2x + 2$ and $3x$, all in cm.

[diagram not to scale]

What is the area, in cm^2, of the triangle?



A 10

B 12

C 28

D 36

E 40

F 54

G 70

WORKED SOLUTION

Using Pythagoras' Theorem, $(3x)^2 = (x+4)^2 + (2x+2)^2$.

Thus

$9x^2 = x^2 + 8x + 16 + 4x^2 + 8x + 4 \Rightarrow 4x^2 - 16x - 20 = 0 \Rightarrow x^2 - 4x - 5$

.

Factorising: $(x-5)(x+1) = 0 \Rightarrow x = -1, 5$. Since $x > 0$, take $x = 5$.

Area = $\frac{1}{2}(x+4)(2x+2) = \frac{1}{2}(9)(12) = 54 \text{ cm}^2$.

M2-11

Mock Test 3 - Mathematics 2

Answer **D**

QUESTION 11

Which of the following is a simplification of $4 - \dfrac{x(3x+1)}{x^2(3x^2-2x-1)}$?

A $\dfrac{12x^3 - 8x^2 - 7x - 1}{x(3x-1)(x-1)}$

B $\dfrac{4x^2 + 4x - 1}{x(x + 1)}$

C $\dfrac{4x^2 + 4x + 1}{x(x + 1)}$

D $\dfrac{4x^2 - 4x - 1}{x(x - 1)}$

E $\dfrac{4x^2 - 4x + 1}{x(x - 1)}$

F $\dfrac{12x^3 - 8x^2 - x + 1}{x(3x - 1)(x - 1)}$

WORKED SOLUTION

In the denominator, $3x^2 - 2x - 1 \equiv (3x + 1)(x - 1)$.

Considering the whole expression: $4 - \dfrac{x(3x + 1)}{x^2(3x + 1)(x - 1)}$.

Cancel $(3x + 1)$: $4 - \dfrac{1}{x(x - 1)}$.

Combine over the common denominator: $\dfrac{4x(x - 1) - 1}{x(x - 1)} = \dfrac{4x^2 - 4x - 1}{x(x - 1)}$.

M2-12 Mock Test 3 - Mathematics 2 Answer B

QUESTION 12

The ball for a garden game is a solid sphere of volume 192 cm^3.

For the children's version of the game the ball is a solid sphere made of the same material, but the radius is reduced by 25%.

What is the volume, in cm^3, of the children's ball?

A 48

B 81

C 96

D 108

E 144

WORKED SOLUTION

$$V = \frac{4}{3}\pi r^3 \Rightarrow V \propto r^3$$

$$V_{child} = \frac{4}{3}\pi r^3_{child}$$

$$= \frac{4}{3}\pi(\frac{3}{4}r)^3 = (\frac{3}{4})^3 V$$

$$= \frac{27}{64} \times 192 = 27 \times 3 = 81 \text{ cm}^3$$

M2-13

Mock Test 3 - Mathematics 2

Answer A

QUESTION 13

Given that

$$9^{2x-1} \times \frac{1}{27^x} = 81^x$$

what is the value of x?

A $-\frac{2}{3}$

B $-\frac{2}{5}$

C $-\frac{1}{3}$

D $-\frac{1}{4}$

E $-\frac{1}{5}$

WORKED SOLUTION

Convert all bases to 3.

$(3^2)^{2x-1} \times \frac{1}{(3^3)^x} = (3^4)^x$

$3^{4x-2} \times 3^{-3x} = 3^{4x}$

$3^{x-2} = 3^{4x}$

Equating exponents: $x - 2 = 4x$

$3x = -2$

$x = -\frac{2}{3}$

M2-14

Mock Test 3 - Mathematics 2

Answer D

QUESTION 14

PR and *QS* are the diagonals of a rhombus *PQRS*.

$PR = (3x + 2)$ cm

$QS = (8 - 2x)$ cm

The area of $PQRS$ is $11\ \mathrm{cm}^2$.

What is the difference, in cm, between the two possible lengths of PR?

A $2\frac{2}{3}$

B $4\frac{1}{2}$

C $5\frac{1}{3}$

D 8

WORKED SOLUTION

Area of rhombus = product of diagonals / 2

$\frac{(3x+2)(8-2x)}{2} = 11$

$12x - 3x^2 + 8 - 2x = 11$

$3x^2 - 10x + 3 = 0$

$(3x - 1)(x - 3) = 0 \Rightarrow x = \frac{1}{3}, 3$

Possible PR lengths: $3(3) + 2 = 11$

$3(\frac{1}{3}) + 2 = 3$

$11 - 3 = 8\,\text{cm}$ difference

M2-15 Mock Test 3 - Mathematics 2 Answer E

QUESTION 15

[diagram not to scale]

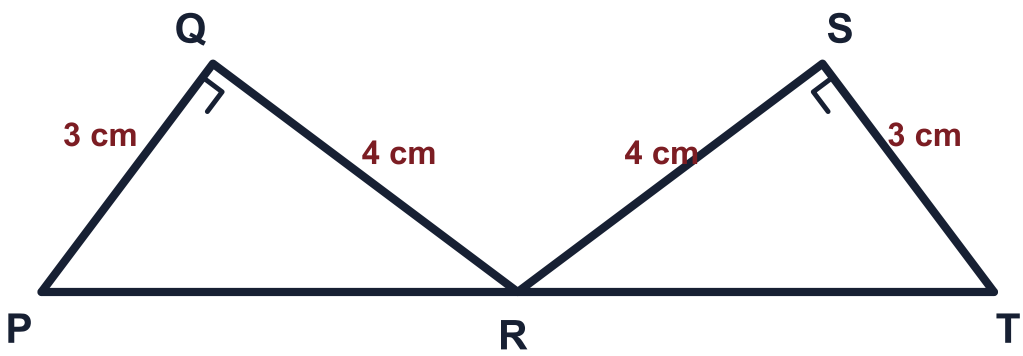
The diagram shows two congruent right-angled triangles PQR and TSR with right angles at Q and S, respectively.

$PQ = TS = 3$ cm

$QR = SR = 4$ cm

PRT is a straight line.

What is the length, in cm, of QS?



A 4

B $3\sqrt{2}$

C 5.2

D $4\sqrt{2}$

E 6.4

WORKED SOLUTION

Notice that the triangle PQR is similar to QUR. This can be deduced by noticing that the angles are the same. The ratio of lengths in PQR is 3:4:5 Therefore QU = 4 x ⅘ = 3.2cm QS = 2 x QU = 6.4cm

M2-16

Mock Test 3 - Mathematics 2

Answer G

QUESTION 16

The total of three numbers p, q and r is 375.

The ratio $p : q$ is $5 : 7$.

The ratio $q : r$ is $4 : 11$.

What is the value of $p + r$?

A 16

B 60

C 97

D 125

E 144

F 231

G 291

H 315

WORKED SOLUTION

q can be written as 75p.

Therefore r can be written as 75p114=7720p

p+q+r=p+75p+7720p=254p=375

Therefore, p=60 and it follows that r=231

60+231=291

M2-17

Mock Test 3 - Mathematics 2

Answer **D**

QUESTION 17

The straight line P has equation $3y - 2x = 12$ and intercepts the y-axis at the point $(0, p)$.

The straight line Q is parallel to P, passes through the point $(6, -1)$ and intercepts the y-axis at the point $(0, q)$.

What is the value of $p - q$?

A −9

B −7

C 1

D 9

E 14

F 17

WORKED SOLUTION

To find the value of p, substitute x = 0 and y = p into 3y - 2x = 12

3p = 12

p = 4

The equation of P can be rearranged to y=23x+4

Therefore the gradient of P and Q is 23

Since Q passes through (6, -1), the equation of Q is:

y-(-1)=23(x-6)

y=23x-5

Substitute x = 0, y = q:

q = -5

p - q = 4 - (-5) = 9

M2-18

Mock Test 3 - Mathematics 2

Answer G

QUESTION 18

The vertices of a rectangle have coordinates:

P(4, 5)   Q(4, 8)   R(10, 8)   S(10, 5)

PQRS is transformed by a clockwise rotation of 90° about P followed by a reflection in the x-axis.

What are the coordinates of the final position of R?

A (-8, -10)

B (-7, -1)

C (-4, 1)

D (-1, 11)

E (1, -11)

F (4, -1)

G (7, 1)

H (8, 10)

WORKED SOLUTION

Relative to $\mathrm{P} = (4, 5)$, point $\mathrm{R} = (10, 8)$ has vector $(6, 3)$.

A clockwise 90° rotation sends $(x, y) \mapsto (y, -x)$, so $(6, 3) \mapsto (3, -6)$.

The rotated image of R is $(4, 5) + (3, -6) = (7, -1)$. Reflecting in the x-axis gives $(7, 1)$.

M2-19

Mock Test 3 - Mathematics 2

Answer C

QUESTION 19

Box A contains exactly 10 balls, of which 6 are red and 4 are blue.

Box B contains exactly 15 balls, of which 3 are red and 12 are blue.

All the balls are identical in every respect, apart from colour.

One of the two boxes is chosen at random by tossing two fair coins, as follows:

"If both coins show heads, box A is selected. Otherwise box B is selected."

One ball is then randomly taken from the selected box.

What is the probability that a red ball is taken?

A $\frac{9}{400}$

B $\frac{3}{25}$

C $\frac{3}{10}$

D $\frac{2}{5}$

E $\frac{1}{2}$

F $\frac{4}{5}$

G $\frac{323}{400}$

WORKED SOLUTION

Box A is selected only if both coins show heads, so $\mathrm{P}(\mathrm{A}) = \frac{1}{4}$ and $\mathrm{P}(\mathrm{B}) = \frac{3}{4}$.

Therefore

$$\mathrm{P}(\mathrm{red}) = \frac{1}{4} \cdot \frac{6}{10} + \frac{3}{4} \cdot \frac{3}{15} = \frac{3}{20} + \frac{3}{20} = \frac{3}{10}.$$

M2-20

Mock Test 3 - Mathematics 2

Answer C

QUESTION 20

Question 20:

A list of five numbers has mean x, median y and range z.

A sixth number is added to the list. This sixth number is greater than x.

Which of the following statements must be true?

- 1 The median of the six numbers cannot be one of the numbers in the list.
- 2 The mean of the six numbers is greater than x.
- 3 The range of the six numbers is greater than z.

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Question 20: C

- Statement 1 is False as the middle two numbers could be equal.
- Statement 2 is True. Since a number larger than the mean is added, the mean will increase.
- Statement 3 is False. For the range to increase, the new number must be the new largest number; this is not necessarily stated.

M2-21 Mock Test 3 - Mathematics 2 Answer C

QUESTION 21

An arithmetic progression has first term a and common difference d.

The sum of the first 5 terms is equal to the sum of the first 8 terms.

Which one of the following expresses the relationship between a and d?

A $a = -\frac{38}{3}d$

B $a = -7d$

C $a = -6d$

D $a = 6d$

E $a = 7d$

F $a = \frac{38}{3}d$

WORKED SOLUTION

One approach is to use the formula for the sum of the first 5 and the first 8 terms:

$$S_n = \frac{1}{2}n(2a + (n - 1)d).$$

We have

$$S_5 = \frac{1}{2} \times 5(2a + 4d) = \frac{5}{2}(2a + 4d),$$

$$S_8 = \frac{1}{2} \times 8(2a + 7d) = 4(2a + 7d).$$

These are equal, so

$$\frac{5}{2}(2a + 4d) = 4(2a + 7d).$$

Multiply both sides by 2 and expand/collect terms:

$$5(2a + 4d) = 8(2a + 7d) \Rightarrow 10a + 20d = 16a + 56d \Rightarrow -6a = 36d \Rightarrow a$$

So the answer is $a = -6d$.

M2-22

Mock Test 3 - Mathematics 2

Answer G

QUESTION 22

Consider the simultaneous equations

$3x^2 + 2xy = 4$

$x + y = a$

where a is a real constant.

Find the complete set of values of a for which the equations have two distinct real solutions for x.

A There are no values of a

B $-2 < a < 2$

C $-1 < a < 1$

D $a = 0$

E $a < -1$ or $a > 1$

F $a < -2$ or $a > 2$

G All real values of a

WORKED SOLUTION

We can solve the equations to find x by substituting for y. The second equation gives $y = a - x$, so the first equation becomes

$$3x^2 + 2x(a - x) = 4.$$

This rearranges to

$$x^2 + 2ax - 4 = 0,$$

whose discriminant is

$$(2a)^2 - 4 \cdot 1 \cdot (-4) = 4a^2 + 16.$$

This is always positive for all real a (since $4a^2 \geq 0$), so there are always two distinct real roots. Hence the correct answer is: all real values of a.

M2-23 Mock Test 3 - Mathematics 2 Answer E

QUESTION 23

Find the shortest distance between the two circles with equations:

$$(x + 2)^2 + (y - 3)^2 = 18$$

$$(x - 7)^2 + (y + 6)^2 = 2$$

A 0

B 4

C 16

D $2\sqrt{2}$

E $5\sqrt{2}$

WORKED SOLUTION

The first circle has centre $(-2, 3)$ and radius $\sqrt{18} = 3\sqrt{2}$.

The second circle has centre $(7, -6)$ and radius $\sqrt{2}$.

It would help to draw a quick sketch of the situation.

The distance between the centres is

$\sqrt{((-2) - 7)^2 + (3 - (-6))^2} = \sqrt{81 + 81} = 9\sqrt{2}$.

Subtracting the two radii ($3\sqrt{2}$ and $\sqrt{2}$) for the parts of the line lying within the circles gives $9\sqrt{2} - 3\sqrt{2} - \sqrt{2} = 5\sqrt{2}$. The correct answer is $5\sqrt{2}$.

This can also be proved by taking an arbitrary point on each circle, finding the distance between them and then minimising this, but that is more complicated than needed.

M2-24 Mock Test 3 - Mathematics 2 Answer D

QUESTION 24

The function f is defined by $f(x) = x^3 + ax^2 + bx + c$.

a, b, c take the values 1, 2 and 3 with no two of them being equal and not necessarily in this order.

The remainder when $f(x)$ is divided by $(x + 2)$ is R.

The remainder when $f(x)$ is divided by $(x+3)$ is S.

What is the largest possible value of $R - S$?

A -26

B 5

C 7

D 17

E 29

WORKED SOLUTION

For this question, we make use of the remainder theorem: the remainder when $f(x)$ is divided by $(x-a)$ is $f(a)$.

In this case, we take $a = -2$ and $a = -3$, so

$f(-2) = R$

$f(-3) = S$

Substituting these into the given formula for $f(x)$ gives

$f(-2) = -8 + 4a - 2b + c = R$

$f(-3) = -27 + 9a - 3b + c = S$

We want the largest possible value of $R - S$, so we subtract these two equations to get

$19 - 5a + b = R - S.$

To get the largest possible value, we want a to be as small as possible and b to be as large as possible, so we take $a = 1, b = 3$ (and then $c = 2$), giving $R - S = 19 - 5 + 3 = 17.$

Thus the correct answer is 17.

M2-25

Mock Test 3 - Mathematics 2

Answer **A**

QUESTION 25

The non-zero constant k is chosen so that the coefficients of x^6 in the expansions of $(1 + kx^2)^7$ and $(k + x)^{10}$ are equal.

What is the value of k?

A $\frac{1}{6}$

B 6

C $\frac{\sqrt{6}}{6}$

D $\sqrt{6}$

E $\frac{\sqrt{30}}{30}$

F $\sqrt{30}$

WORKED SOLUTION

Question 25: A

We can expand these two expressions using the binomial theorem, focusing on the term involving x^6:

$$(1 + kx^2)^7 = \cdots + \binom{7}{3} 1^4 (kx^2)^3 + \cdots = \cdots + \frac{7\times 6\times 5}{3\times 2\times 1} k^3 x^6 + \cdots = \cdots + 35k^3$$

and

$$(k + x)^{10} = \cdots + \binom{10}{6} k^4 x^6 + \cdots = \cdots + \frac{10\times 9\times 8\times 7}{4\times 3\times 2\times 1} k^4 x^6 + \cdots = \cdots + 210k^4$$

We therefore need $35k^3 = 210k^4$, so $k = \frac{35}{210} = \frac{5}{30} = \frac{1}{6}$.

Thus the correct answer is $\frac{1}{6}$.

M2-26

Mock Test 3 - Mathematics 2

Answer B

QUESTION 26

Find the complete set of values of the constant c for which the cubic equation

$$2x^3 - 3x^2 - 12x + c = 0$$

has three distinct real solutions.

A $-20 < c < 7$

B $-7 < c < 20$

C $c > 7$

D $c > -7$

E $c < 20$

F $c < -20$

WORKED SOLUTION

Question 26: B

Sketch the cubic by finding its turning points and the y-intercept. For $y = 2x^3 - 3x^2 - 12x + c$, differentiate:

$\frac{dy}{dx} = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x - 2)(x + 1)$.

Hence the stationary points are at $x = 2$ and $x = -1$ with coordinates $(-1, 7 + c)$ and $(2, -20 + c)$.

For $2x^3 - 3x^2 - 12x + c = 0$ to have three distinct real solutions, the turning points must lie on opposite sides of the x-axis. Since $-20 + c < 7 + c$, we need

$-20 + c < 0$ and $7 + c > 0$.

Thus $c < 20$ and $c > -7$. Combining gives $-7 < c < 20$.

M2-27

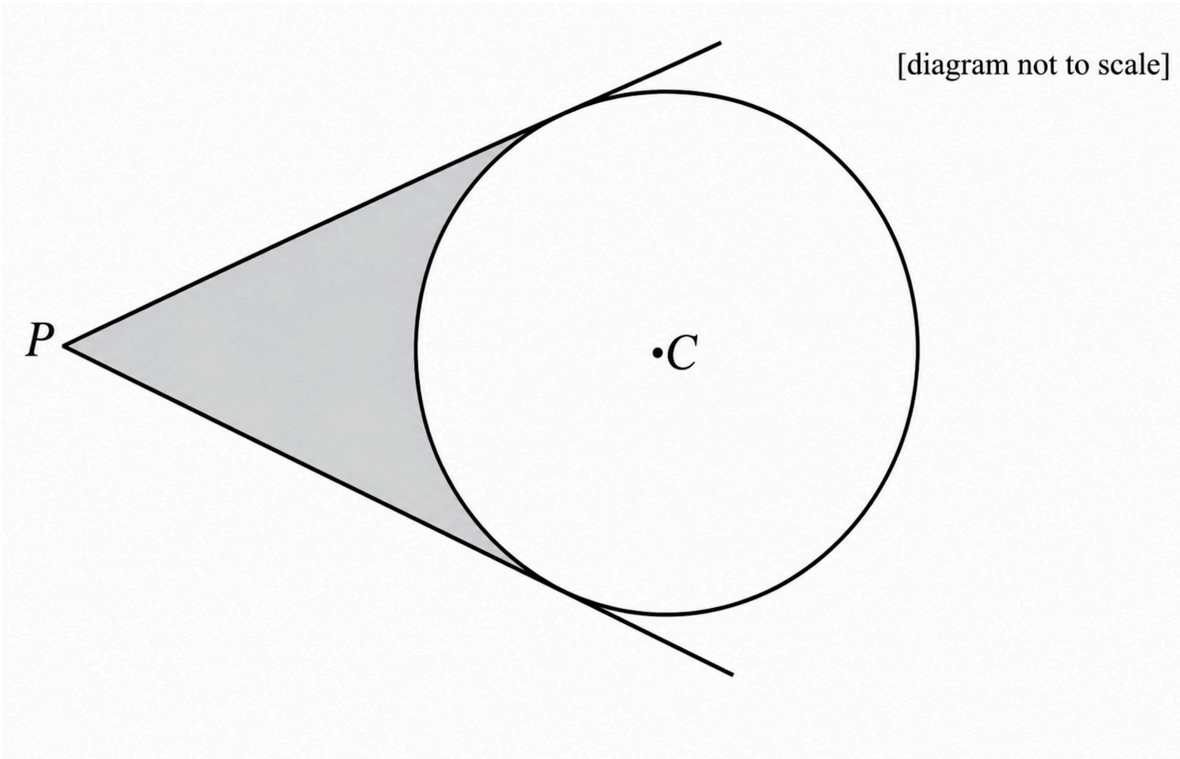
Mock Test 3 - Mathematics 2

Answer **A**

QUESTION 27

Tangents are drawn from a point P to a circle of radius 10 cm. The centre of the circle is C and the distance PC is 20 cm.

Which one of the following is an expression for the shaded area in square centimetres?



A $\frac{100}{3}(3\sqrt{3} - \pi)$

B $\frac{100}{3}(3\sqrt{5} - \pi)$

C $\frac{50}{3}(6\sqrt{3} - \pi)$

D $\frac{50}{3}(6\sqrt{5} - \pi)$

E $\frac{50}{3}(\sqrt{3} - 2\pi)$

WORKED SOLUTION

sin $\theta = 10/x = 10/20 = 1/2$

$\theta = 30^\circ$

Area of the triangle:

$2 \times (\frac{1}{2} \times x \cos\theta \times 10)$

= $x\sqrt{3}/2 \times 10$

= $5x\sqrt{3}$

= $100\sqrt{3}$

Area of circular sector within triangles = $120/360 \times \pi \times 10^2$

= $100/3\pi$

Therefore shaded area = $100\sqrt{3} - 100/3\pi$

= $\frac{100}{3}(3\sqrt{3} - \pi)$

LEVEL 1 · STANDARD

Biology

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Biology Standard source set. Question wording, option order, answer and solution method are unchanged.

B-01

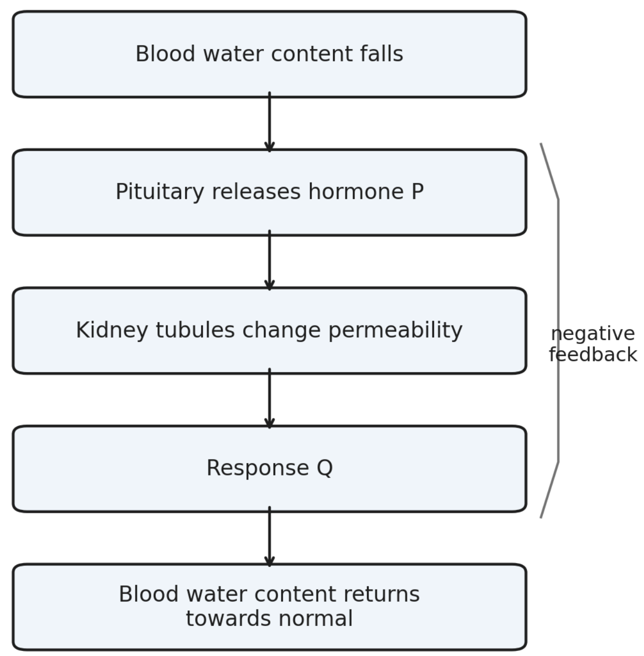
Homeostasis

Answer D

QUESTION 1

A person loses water by sweating for a prolonged period.

The diagram summarises part of the homeostatic response.



A insulin; more water reabsorbed by the kidneys

B glucagon; less water reabsorbed by the kidneys

C ADH; less water reabsorbed and a larger volume of urine produced

D ADH; more water reabsorbed and a smaller volume of urine produced

E adrenaline; less water reabsorbed and more concentrated urine

WORKED SOLUTION

Low blood water content causes increased release of ADH from the pituitary gland.

ADH increases the permeability of parts of the kidney tubule, so more water is reabsorbed into the blood. Less water is therefore lost in urine, giving a smaller volume of more concentrated urine.

Hence the correct answer is **D**.

B-02

Cell structure | Answer **C**

QUESTION 2

A cell has:

- a cell membrane;
- cytoplasm;
- a cell wall;
- circular chromosomal DNA;
- several small plasmids;
- no mitochondria.

Which statement about the cell is correct?

A It must contain chloroplasts.

B It cannot carry genetic information.

C A plasmid may carry a gene for antibiotic resistance.

D It is a eukaryotic cell.

E It cannot carry out respiration.

WORKED SOLUTION

The combination of circular chromosomal DNA, plasmids and absence of mitochondria is characteristic of a bacterial cell.

Plasmids can carry additional genes, including genes that confer antibiotic resistance. Bacteria can still carry out respiration even though they do not possess mitochondria.

Therefore the correct answer is **C**.

B-03 DNA and base composition Answer C

QUESTION 3

A double-stranded DNA molecule is **300 base pairs** long.

Adenine accounts for 36% of all bases in the molecule.

How many guanine bases are present?

A 42

B 72

C 84

D 108

E 168

WORKED SOLUTION

There are

$$300 \times 2 = 600$$

bases in total.

In double-stranded DNA, adenine and thymine occur in equal proportions. Therefore

$$A + T = 36\% + 36\% = 72\%.$$

The remaining 28% consists equally of cytosine and guanine, so guanine makes up 14%.

$$0.14 \times 600 = 84.$$

Thus the correct answer is **C**.

B-04

Natural selection

Answer G

QUESTION 4

A population of beetles contains both pale and dark individuals.

Before an industrial development, most tree bark in the beetles' habitat is pale. After many years of soot deposition, most bark becomes much darker. The frequency of an allele associated with dark colour subsequently increases.

Which statements could explain the change?

1. Soot caused pale beetles to mutate specifically into dark beetles.
2. Dark beetles may have become less visible to predators.
3. If colour is heritable, dark beetles surviving and reproducing more successfully could increase the frequency of the dark allele.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Mutation is not directed by environmental need, so statement 1 is incorrect.

Dark beetles could be better camouflaged on darkened bark, increasing their chance of survival. If the difference is heritable, greater reproductive success of dark beetles can increase the frequency of the associated allele.

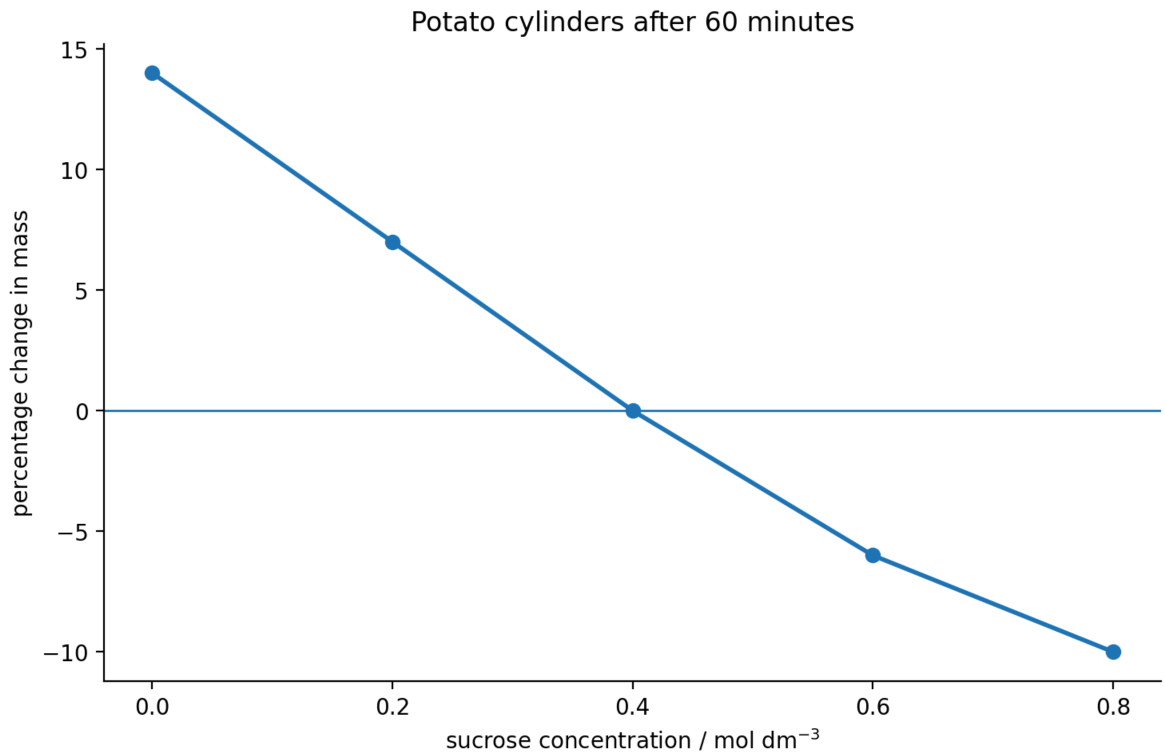
Statements 2 and 3 are therefore correct, giving **G**.

B-05 Osmosis Answer **C**

QUESTION 5

Identical potato cylinders were placed for 60 minutes in different sucrose solutions.

Which row is correct?



	concentration giving approximately no net change in mass / mol dm^{-3}	net movement of water at 0.20 mol dm^{-3}
A	0.20	into the potato cells
B	0.40	out of the potato cells
C	0.40	into the potato cells
D	0.60	into the potato cells
E	0.40	no net movement

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

At approximately $0.40\ \mathrm{mol\ dm^{-3}}$, the potato cylinders show no percentage change in mass, indicating no net movement of water.

At $0.20\ \mathrm{mol\ dm^{-3}}$, the cylinders gain mass, so there has been a net movement of water into the potato cells by osmosis.

Therefore row **C** is correct.

B-06

Enzymes

Answer **F**

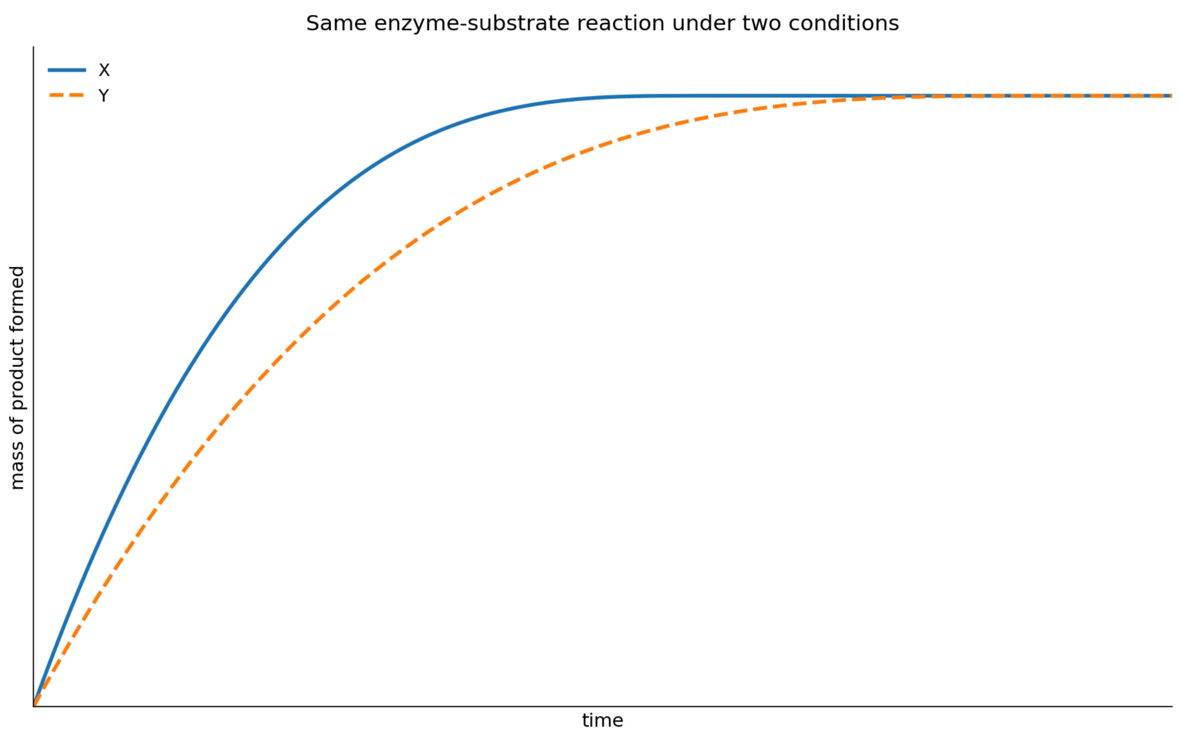
QUESTION 6

The graph shows the mass of product produced during the same enzyme-controlled reaction under two different conditions, X and Y.

The graph shows both reactions until no further product is formed.

Which changes **could** explain the difference between X and Y?

1. X contains a greater concentration of enzyme than Y.
2. X contains a greater initial amount of substrate than Y.
3. The temperature in X is closer to the enzyme's optimum than the temperature in Y, with neither temperature causing denaturation.



A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Both curves reach the same final mass of product, so the graph does not support X having a greater initial amount of substrate than Y.

A greater enzyme concentration can increase the rate of reaction without changing the total amount of substrate ultimately converted. A temperature closer to the enzyme's optimum can also increase the rate, provided neither temperature causes denaturation.

Therefore statements 1 and 3 could explain the graph: **F**.

B-07 Cell division Answer **E**

QUESTION 7

A human body cell contains 46 chromosomes and $6\ \mathrm{pg}$ of DNA before DNA replication.

Which row correctly describes the same cell immediately before mitosis and one daughter cell after mitosis?

	immediately before mitosis: chromosomes	immediately before mitosis: DNA / pg	one daughter cell after mitosis: chromosomes	one daughter cell after mitosis: DNA / pg
A	23	12	23	6
B	46	6	46	6
C	92	12	46	6
D	46	12	23	6
E	46	12	46	6

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

DNA replication doubles the amount of DNA from 6 pg to 12 pg, but the chromosome number is still counted as 46 because each chromosome consists of two sister chromatids.

After mitosis, each daughter cell has the normal diploid chromosome number and half the replicated DNA:

$$46 \text{ chromosomes}, \qquad 6 \text{ pg DNA}.$$

So row **E** is correct.

B-08

Stem cells

Answer F

QUESTION 8

Which statements about an adult stem cell from human bone marrow are correct?

1. It may divide by mitosis to produce more stem cells.
2. It is totipotent and can form an entire new human organism.
3. It can differentiate into a limited range of specialised blood cells.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Adult stem cells can divide by mitosis and self-renew. Bone-marrow stem cells are multipotent: they can form a restricted range of specialised blood cells.

They are not totipotent and cannot normally form an entire new organism.

Therefore statements 1 and 3 only are correct: **F**.

B-09

Inheritance and pedigrees

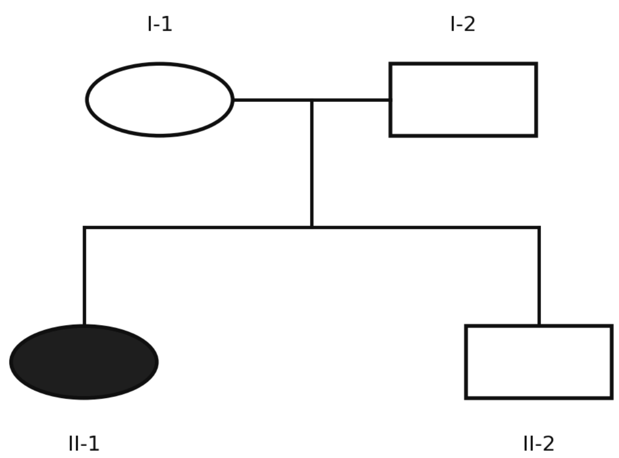
Answer C

QUESTION 9

The condition shown in the pedigree is caused by an **autosomal recessive allele**.

Individual II-2 is unaffected.

If II-2 later has a child with a woman who is affected by the condition, what is the probability that their child will be affected?



Filled symbol = affected by an autosomal recessive condition

A $\frac{1}{6}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{1}{2}$

E $\frac{2}{3}$

WORKED SOLUTION

Because two unaffected parents produced an affected child with a recessive condition, both parents must be heterozygous:

$$\mathrm{Aa} \times \mathrm{Aa}.$$

Given that II-2 is unaffected, the possible genotypes are $\mathrm{AA}, \mathrm{Aa}, \mathrm{Aa}$. Therefore

$$\mathrm{P}\left(\text{II-2 is Aa}\right) = \frac{2}{3}.$$

The affected woman is aa. If II-2 is a carrier, the probability that their child is affected is $\frac{1}{2}$.

Hence

$$\frac{2}{3} \times \frac{1}{2} = \boxed{\frac{1}{3}}.$$

The correct answer is **C**.

B-10

Monohybrid inheritance

Answer **D**

QUESTION 10

In rabbits, black fur is caused by a dominant allele B, while brown fur is caused by the recessive allele b.

Two black rabbits produce a brown offspring.

What is the probability that their **next two offspring are both black**?

A $\frac{1}{4}$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{9}{16}$

E $\frac{3}{4}$

WORKED SOLUTION

A brown offspring must have genotype bb, so each black parent must have contributed a recessive allele. The parental cross is therefore

$$Bb \times Bb.$$

The probability that one offspring is black is $\frac{3}{4}$.

For two successive offspring:

$$\frac{3}{4} \times \frac{3}{4} = \boxed{\frac{9}{16}}.$$

Therefore the answer is **D**.

B-11 DNA and mutation Answer **B**

QUESTION 11

The relevant coding-DNA triplets are shown.

A section of coding DNA is

$$\text{ATG CCT GAA TTT}.$$

A substitution changes the second triplet from CCT to CAT.

Which statements are correct?

1. Exactly one amino acid in the section is changed.
2. The reading frame is shifted.
3. The number of amino acids coded by the shown section must decrease.

coding-DNA triplet	amino acid
ATG	methionine
CCT	proline
CAT	histidine
GAA	glutamate
TTT	phenylalanine

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The substitution changes the triplet from CCT to CAT, which changes proline to histidine.

A substitution neither inserts nor removes a base, so the reading frame is not shifted. No information given indicates that a stop signal has been introduced.

Thus only statement 1 is correct: **B**.

B-12

Gene technology

Answer E

QUESTION 12

A suitable copy of a human gene is inserted into a bacterial plasmid so that bacteria can produce the corresponding human protein.

Which statements are correct?

1. A restriction enzyme may be used to cut DNA.
2. DNA ligase may be used to join the human gene to plasmid DNA.
3. The recombinant plasmid must enter the nucleus of the bacterial cell before the gene can function.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Restriction enzymes can cut DNA at particular sequences, and DNA ligase can join the suitable human gene copy to plasmid DNA.

Bacteria do not possess a nucleus, so the recombinant plasmid does not have to enter one.

Therefore statements 1 and 2 only are correct: **E**.

B-13

Respiration during exercise

Answer **B**

QUESTION 13

The table shows oxygen demand and oxygen supplied to working muscle at different times during exercise.

Here, oxygen demand means the amount of oxygen that would be required to meet the muscle's energy demand entirely by aerobic respiration.

Assume the oxygen supplied is all available for aerobic respiration.

At which measured times must anaerobic respiration also be contributing to the muscle's energy supply?

time / min	oxygen demand / arbitrary units	oxygen supplied / arbitrary units
0	0.8	0.8
2	1.4	1.0
4	1.8	1.3
6	2.0	1.7
8	1.6	1.6
10	1.1	1.2

- **A** 0 only
- **B** 2, 4 and 6 only
- **C** 8 only
- **D** 2, 4, 6 and 8
- **E** 10 only

WORKED SOLUTION

When this oxygen demand exceeds the oxygen supplied, aerobic respiration alone cannot meet the muscle's stated energy demand, so anaerobic respiration must also contribute.

This occurs at 2, 4 and 6 minutes.

At 8 minutes demand equals supply, and at 10 minutes supply is greater than demand.

Therefore the answer is **B**.

B-14 Circulation Answer **D**

QUESTION 14

Three blood vessels have the following properties.

- Vessel P carries blood **from the heart to the lungs** and has relatively low oxygen concentration.
- Vessel Q carries blood **from the lungs to the heart** and has relatively high oxygen concentration.
- Vessel R carries oxygenated blood **from the heart to a kidney**.

Which row correctly identifies the vessels?

	P	Q	R
A	pulmonary vein	pulmonary artery	renal vein
B	aorta	vena cava	pulmonary artery
C	pulmonary artery	pulmonary vein	renal vein
D	pulmonary artery	pulmonary vein	renal artery
E	vena cava	pulmonary artery	aorta

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The pulmonary artery carries relatively deoxygenated blood from the heart to the lungs.

The pulmonary vein carries oxygenated blood from the lungs to the heart.

The renal artery carries oxygenated blood from the heart to a kidney.

Therefore row **D** is correct.

B-15 Digestion Answer **C**

QUESTION 15

Four tubes are incubated at $37^{\circ}C$.

The pH of each tube is monitored continuously.

The lipase used has an optimum near pH 8. Bile emulsifies lipid into smaller droplets but contains no digestive enzyme.

Which tube would be expected to show the **greatest initial rate of decrease in pH**?

tube	contents
A	lipid + water
B	lipid + lipase at pH 8
C	lipid + lipase + bile at pH 8
D	lipid + lipase + bile at pH 2

A A

B B

C C

D D

E All at the same rate

WORKED SOLUTION

Lipase hydrolyses lipids to products including fatty acids, so lipase activity causes the pH to fall.

Tubes B and C are at pH 8, near the stated optimum for the lipase. Tube C also contains bile, which emulsifies the lipid into smaller droplets and increases the surface area available to the enzyme. Tube D is at pH 2, far from the stated optimum, and tube A contains no lipase.

Tube C therefore has the conditions expected to give the greatest initial rate of decrease in pH. The answer is **C**.

B-16

Kidney function

Answer E

QUESTION 16

The table gives relative concentrations of substances in three fluids from a healthy person.

Which statements are correct?

1. Most plasma proteins are prevented from entering the filtrate.
2. Glucose is reabsorbed from the nephron.
3. The greater concentration of urea in urine proves that urea is produced by kidney cells.

substance	blood plasma	filtrate entering nephron	urine
protein	8.0	0	0
glucose	1.0	1.0	0
urea	0.3	0.3	2.0

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The absence of protein from the filtrate is consistent with large plasma proteins being retained in the blood.

Glucose is present in the filtrate but absent from normal urine, supporting its reabsorption.

A greater urea concentration in urine does not show that the kidney produces urea; water removal can concentrate urea already present.

Therefore statements 1 and 2 only are correct: **E**.

B-17

Nervous system

Answer **E**

QUESTION 17

Which statements about a simple reflex arc are correct?

1. A sensory neurone can carry an impulse from a receptor towards the central nervous system.
2. A relay neurone can connect a sensory neurone to a motor neurone within the central nervous system.
3. A motor neurone normally carries impulses from an effector towards the central nervous system.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Sensory neurones carry impulses from receptors towards the CNS. Relay neurones can connect sensory and motor neurones within the CNS.

Motor neurones carry impulses from the CNS to effectors, not from effectors back towards the CNS.

Therefore statements 1 and 2 only are correct: **E**.

B-18 Disease and body defence Answer **H**

QUESTION 18

Which statements are correct?

1. Antibiotics do not directly destroy viruses.
2. Vaccination may lead to the formation of memory cells.
3. Antibiotic treatment can favour the survival and reproduction of resistant bacteria already present in a population.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

All three statements are correct.

Antibiotics target bacteria rather than directly destroying viruses. Vaccination can produce memory cells. Antibiotic exposure also acts as a selection pressure, allowing resistant bacteria to survive and reproduce more successfully.

Therefore the answer is **H**.

B-19 Ecosystems Answer **B**

QUESTION 19

Students record four species along a transect.

Which statements are correct?

1. Species richness is greatest at both 10 m and 20 m.
2. Total abundance is greatest at 30 m.
3. Species C occurs at every sampled position.

distance / m	species A	species B	species C	species D
0	4	0	2	0
10	3	1	2	0
20	0	4	3	1
30	0	2	0	5

A none

- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Species richness is the number of different species present. At both 10 m and 20 m, three species are present, the greatest richness recorded.

Total abundance is $6, 6, 8, 7$ at 0, 10, 20 and 30 m respectively, so statement 2 is false.

Species C is absent at 30 m, so statement 3 is false.

Thus statement 1 only is correct: **B**.

B-20 Reproductive hormones Answer **C**

QUESTION 20

Hormones P, Q and R have the following roles.

- P is released from the pituitary gland and stimulates development of an ovarian follicle.
- Q is released from the pituitary gland and its rise triggers ovulation.
- R is released from the ovary after ovulation and helps maintain the uterine lining.

Which row correctly identifies P, Q and R?

	P	Q	R
A	FSH	progesterone	LH
B	LH	FSH	progesterone
C	FSH	LH	progesterone
D	oestrogen	LH	FSH
E	FSH	LH	oestrogen

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

FSH stimulates development of an ovarian follicle. LH triggers ovulation. Progesterone is produced after ovulation and helps maintain the uterine lining.

Therefore the correct row is **C**.

B-21 Photosynthesis and limiting factors Answer **E**

QUESTION 21

A plant is exposed to different light intensities and carbon dioxide concentrations. All other conditions are kept constant.

Which statements are supported by the data?

1. At low light intensity, light could be the limiting factor.
2. At high light intensity and 0.04% carbon dioxide, carbon dioxide could be a limiting factor.
3. Increasing light intensity and carbon dioxide concentration indefinitely must continue increasing the rate.

light intensity	carbon dioxide concentration	rate of photosynthesis / arbitrary units
low	0.04%	5
low	0.10%	5
high	0.04%	8
high	0.10%	14

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

At low light intensity, increasing carbon dioxide does not increase the rate, so light could be limiting.

At high light intensity, increasing carbon dioxide from 0.04% to 0.10% increases the rate, so carbon dioxide could be limiting at 0.04%.

The rate cannot be assumed to increase indefinitely because another factor will eventually become limiting.

Therefore statements 1 and 2 only are correct: **E**.

B-22

Transpiration

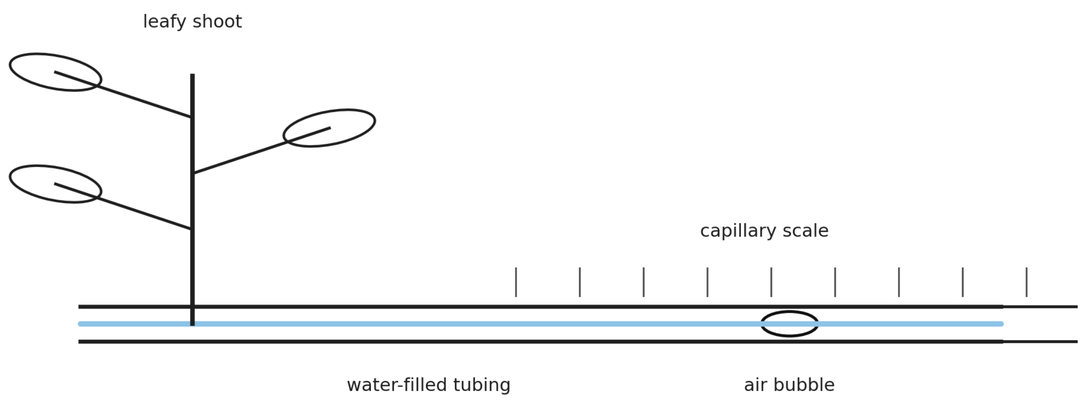
Answer **D**

QUESTION 22

A potometer is used to estimate water uptake by a leafy shoot.

The air bubble moves $18\ \mathrm{mm}$ in 6 minutes. The cross-sectional area of the capillary tube is $0.50\ \mathrm{mm}^2$.

Which row gives the rate of water uptake and the likely effect of increasing the humidity around the leaves?



	water uptake / $mm^3\ min^{-1}$	movement of bubble when humidity increases
A	0.75	faster
B	0.75	slower
C	1.50	faster
D	1.50	slower
E	3.00	slower

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The volume of water taken up is

$$18 \times 0.50 = 9.0\ \text{mm}^3.$$

Over 6 minutes, the rate is

$$\frac{9.0}{6} = 1.50\ \text{mm}^3\ \text{min}^{-1}.$$

Increasing humidity reduces the water-vapour concentration gradient between the leaf and surrounding air, so transpiration decreases and the bubble moves more slowly.

Therefore row **D** is correct.

B-23

Translocation

Answer B

QUESTION 23

A plant is exposed to carbon dioxide containing radioactive carbon.

Several hours later, radioactive sucrose is detected in the roots.

Which explanation is correct?

- **A** Radioactive carbon dioxide entered the roots and was converted directly to sucrose.
- **B** Radioactive carbon dioxide entered photosynthetic tissue, was incorporated into sugars, and some sugar was transported to the roots through the phloem.
- **C** Radioactive carbon dioxide entered the leaves and was transported to the roots through the xylem before photosynthesis occurred.
- **D** Radioactive water formed in the leaves and moved through the phloem.
- **E** Radioactive carbon dioxide was absorbed directly by root hair cells by active transport.

WORKED SOLUTION

Carbon dioxide enters photosynthetic tissue and is incorporated into organic compounds during photosynthesis.

Sugars such as sucrose can then move from source tissues such as leaves to sink tissues such as roots by translocation through the phloem.

Therefore the correct answer is **B**.

B-24

Root uptake

Answer B

QUESTION 24

Root hair cells contain a higher concentration of nitrate ions than the surrounding soil solution.

Nitrate ions nevertheless continue to enter the cells.

When a chemical that inhibits respiration is added, nitrate uptake falls sharply, but water can initially continue to enter the cells.

Which row best explains these observations?

	nitrate-ion uptake	water uptake
A	diffusion	active transport
B	active transport	osmosis
C	diffusion	diffusion
D	osmosis	diffusion
E	active transport	active transport

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

Nitrate ions are moving into cells despite already being at a higher concentration inside, so uptake is against the concentration gradient and requires active transport.

Respiration provides energy for this process, explaining why the inhibitor reduces nitrate uptake.

Water enters cells across a partially permeable membrane by osmosis.

Therefore row **B** is correct.

B-25

Surface area to volume ratio

Answer **C**

QUESTION 25

A cube of living tissue has sides of 3 mm.

It is cut into 27 separate cubes, each with sides of 1 mm. The total volume of tissue is unchanged.

Which row correctly describes the effect?

	total surface area after cutting compared with before	**maximum diffusion distance to the centre of a cube**
A	unchanged	decreases
B	doubles	unchanged
C	triples	decreases
D	increases sixfold	increases
E	increases ninefold	decreases

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The original surface area is

$$6(3^2) = 54\ \mathrm{mm}^2.$$

Each $1\ \mathrm{mm}$ cube has surface area $6\ \mathrm{mm}^2$, so 27 cubes have total surface area

$$27(6) = 162\ \mathrm{mm}^2.$$

Thus surface area triples:

$$\frac{162}{54} = 3.$$

The maximum distance from the surface to the centre falls from $1.5\ \mathrm{mm}$ to $0.5\ \mathrm{mm}$.

Therefore row **C** is correct.

B-26 Meiosis and sex determination Answer **F**

QUESTION 26

A cell in the testes of a healthy human male undergoes meiosis to produce sperm.

Which statements are correct?

1. Each normal sperm produced contains 23 chromosomes.

2. Every sperm produced contains a Y chromosome.

3. Fertilisation of an egg by a Y-bearing sperm can produce an XY zygote.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Meiosis produces haploid gametes containing 23 chromosomes.

Sperm contain either an X chromosome or a Y chromosome, so statement 2 is false.

A Y-bearing sperm fertilising an X-bearing egg can produce an XY zygote.

Therefore statements 1 and 3 only are correct: **F**.

B-27

Blood glucose control

Answer **E**

QUESTION 27

After a carbohydrate-rich meal, the blood glucose concentration of a healthy person rises.

Which statements are correct?

1. Insulin secretion should increase.
2. Increased uptake of glucose by target tissues can contribute to the return of blood glucose concentration towards its normal level.
3. Glucagon secretion should increase because glucagon promotes uptake of glucose from the blood into the liver.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A rise in blood glucose concentration stimulates increased insulin secretion.

Insulin promotes processes that remove glucose from the blood, helping return the concentration towards its normal range by negative feedback.

Glucagon acts in the opposite direction when blood glucose is low; it promotes processes that increase blood glucose concentration.

Therefore statements 1 and 2 only are correct: **E**.

Compilation record: Level 1 uses Engineering Mock Test 3 for Mathematics/Physics and the corresponding Standard Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.